



INNOVATION

IN EVERY SPHERE OF LIFE

JANANI GOPALAKRISHNAN VIKRAM



A smart pan with built-in sensors and an app that helps you cook perfectly all the time (Courtesy: Smartpans)

Smart pan that lets you cook perfectly

Computer engineer Rahul Baxi and his nutritionist sister Prachi put their heads together to develop **Smartpans**—a smart pan that makes anybody a wonderful cook!

The pan is smart in many ways. First, even an amateur can whip up amazing dishes with this pan and its associated app. You can pick recipes by master chefs or from the community and start cooking. The app will give you instructions on what to add, what to do next and when to turn the pan off! The built-in weight and temperature sensors make sure you do not add too much oil, char the dish or commit other such silly mistakes that could spoil the dish. While the weight sensors keep measuring and showing you each of the ingredients you add to the pan, the temperature sensor keeps track of the cooking temperature and modifies each recipe based on your stove's temperature.

You can also share your recipes with your friends instantly. As you cook, just tell the app what ingredients you are adding. The app records the weight of each ingredient, the temperature and the cooking time. Click a pic at the end and voila, your recipe is ready to share.

The pan has a unique lighting system for notifications. It works on a 2500mAh rechargeable Li-Po battery that lasts over four months on a single charge. With a nice, double-layered heavy bottom, the pan is also durable. The colourful, detachable cooking inserts come in ceramic non-stick and stainless steel.

The Smartpans app also syncs with fitness trackers like Fitbit. As and when you cook, the app computes the nutritional value of the dish and shares it with your fitness trackers at the click of a button. So, now you just cannot go wrong with this pan!

Company: Smartpans; **Country:** USA; **Website:** <https://smartpans.io>

MEMS microphone to voice-activate battery-powered devices

Imagine activating your television remote and switching on the TV with just a voice command instead of walking over to it? However, speech capabilities consume a lot of power, often 1000µW or more, which makes them impractical for battery-powered devices like television remotes. Now, **Vesper's VM1010 ZeroPower Listening (ZPL) microphone** can help you voice-activate such devices too.

The VM1010 is a piezoelectric micro-electro-mechanical-system (MEMS) microphone with a tiny footprint of 3.76x2.65x0.96mm³. When the environment is silent, the device stays in the listening or wake-on-sound mode, which consumes just 6µA, while in normal mode it uses around 90µA. The microphone simply uses sound energy and the power that is automatically dissipated by batteries, thus practically consuming no power for the device it is used in.



This MEMS microphone uses sound energy to wake up battery-powered devices (Courtesy: Vesper)

Using special materials, the microphone converts sound energy directly into electrical energy to wake up devices from sleep. According to the company, the device uses piezoelectric effect to make the acoustic transducer operate as an acoustic switch. When a sound wave hits a piezoelectric cantilever, it moves the cantilever. This motion creates a voltage via the piezoelectric effect. This voltage is sensed by a very low-power comparator circuit, which sends a wake signal to the rest of the system. All sounds create a voltage on the cantilever but the circuit responds only to the human vocal range (sounds between 300Hz and 6kHz). Users can program such aspects as loud-

ness threshold, desired distance and level of background noise.

Company: Vesper; **Country:** USA; **Website:** <http://vespermems.com/products/vm1010>

Keep fit with this Wi-Fi smart weighing scale

Fitness trackers help you stay fit but their effectiveness also depends on whether you are able to provide accurate and regular updates on your food intake, weight and other parameters. The more the inputs are automated, the better the performance of fitness trackers. Fitbit's Aria falls in line with this concept.



This Wi-Fi smart scale syncs with your Fitbit app to give you the real picture and help you attain your fitness goals (Courtesy: Fitbit)

Aria is a sleek weighing scale with a polished glass surface that tracks your weight, body mass index (BMI), lean mass and body fat percentage. It recognises up to eight users and keeps each one's recordings confidential.

As soon as you step on the scale, it recognises you, displays your readings and sends it across to your Fitbit

account using your home Wi-Fi network. On your Fitbit app, you can see weight statistics and progress trends in the form of easy-to-read charts and graphs. This is very important because long-term weight trends are better indicators of health and fitness than short-term fluctuations.

With the ability to record workouts and log food habits and weight too, the Fitbit app becomes a one-stop solution for your fitness needs. You can make the weighing scale, the tracker and the app work in sync by setting goals and using calorie coaching to achieve them.

You can buy Aria at the Fitbit store on Amazon.in.

Company: Fitbit; **Country:** USA; **Website:** <https://www.fitbit.com/in/aria>